Data introduction: EEG and MRI

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870
ADNI MRI subjects
294 AD · 576 CN

67
BrainLat EEG subjects
35 AD · 32 CN

65
ds004504 EEG subjects
36 AD · 29 CN

1. Overview

The project compares Alzheimer's disease (AD) with cognitively normal controls (CN) in two modalities. Structural MRI comes from ADNI as FreeSurfer-derived regional measures (no images are used yet). Resting-state EEG comes from two independent public datasets, BrainLat (Latin America, two sites) and OpenNeuro ds004504 (Greece, one site), processed with the same pipeline. The labels are consistent across all three: **dx = AD** / **CN** , **sex = M** / **F** .

Dataset	Modality	Data form	Location	Size used	AD	CN
ADNI (UCSF FreeSurfer 7, UCSFFSX7)	Structural MRI, 3T T1	Tabular FreeSurfer measures (324 per subject)	/project/mlresearch/Runkun/MRI → processed/mri	82 MB	294	576
BrainLat (Synapse syn51549340)	Resting-state EEG	EEGLAB .set/.fdt, 128-ch BioSemi, 512 Hz	/projectnb/mlresearch/Runkun/brainlat_eeg_full	8.0 GB	35	32
OpenNeuro ds004504	Resting-state EEG, eyes closed	BIDS, EEGLAB .set, 19-ch 10–20, 500 Hz	/projectnb/mlresearch/Runkun/ds004504	2.2 GB	36	29

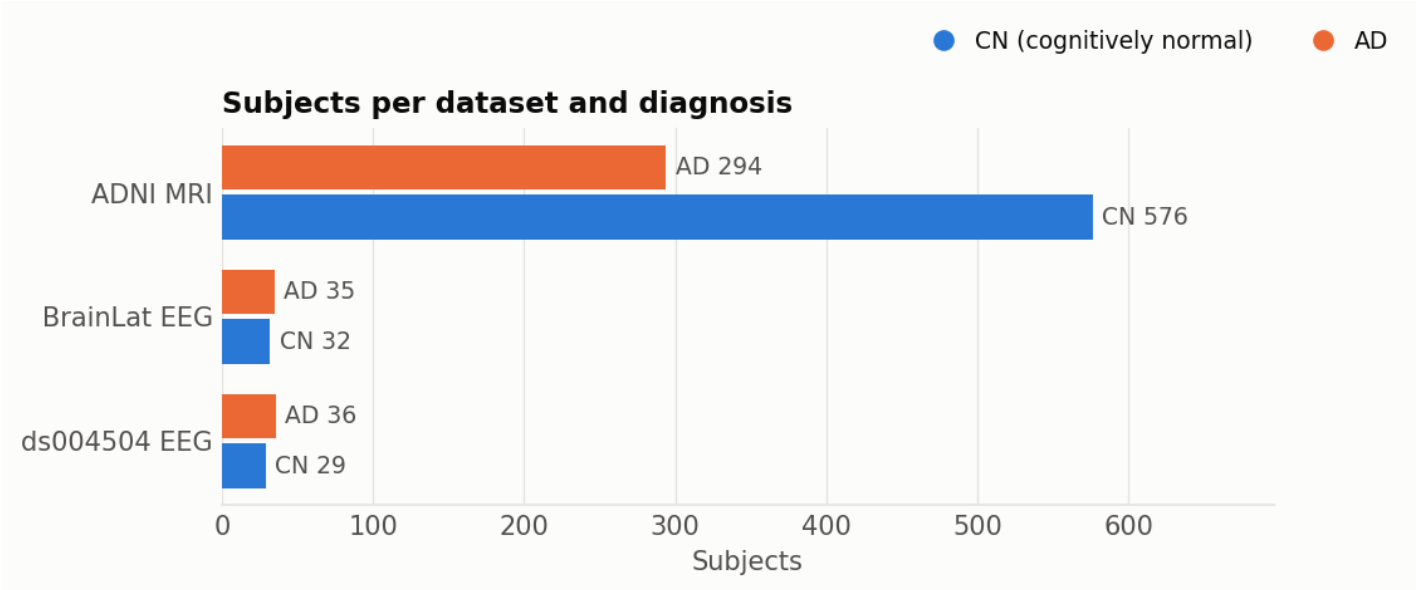


Figure 1. Subjects per dataset. ADNI CN subjects are age- and sex-matched 2:1 to AD.

How each cohort was selected

Dataset	Step	AD	CN
BrainLat	Subjects in Records CSVs	35	46
BrainLat	Signal file (.fdt / embedded data) on Synapse	35	32
BrainLat	Loads, harmonized, features extracted	35	32
ds004504	Participants (Group A / C; 23 FTD excluded)	36	29
ds004504	Loads, harmonized, features extracted	36	29
ADNI MRI	AD/CN baseline 3T scans, CN age/sex-matched 2:1	294	576

BrainLat: 14 HC recordings are header-only on Synapse (the .fdt signal file does not exist anywhere in the project), so CN drops from 46 to 32. All 35 AD recordings are usable.

2. Demographics

Dataset	Group	n	Age, mean (SD)	Female, n (%)	Education, y	MMSE	p (age)	p (sex)	p (education)
ADNI MRI	CN	576	73.3 (7.8)	290 (50%)	16.6 (2.5)	–			
ADNI MRI	AD	294	73.2 (7.6)	146 (50%)	15.5 (2.8)	–	0.961	0.904	<0.001
BrainLat EEG	CN	32	69.8 (7.7)	20 (62%)	15.6 (3.4)	–			
BrainLat EEG	AD	35	76.5 (7.6)	22 (63%)	9.3 (4.6)	–	0.002	1.000	<0.001
ds004504 EEG	CN	29	67.9 (5.4)	11 (38%)	not recorded	30.0 (0.0)			
ds004504 EEG	AD	36	66.4 (7.9)	24 (67%)	not recorded	17.8 (4.5)	0.383	0.039	–

p values: Mann–Whitney U for age and education, χ^2 for sex (AD vs CN within each dataset). MMSE is only available for ds004504, where every CN subject scores 30.

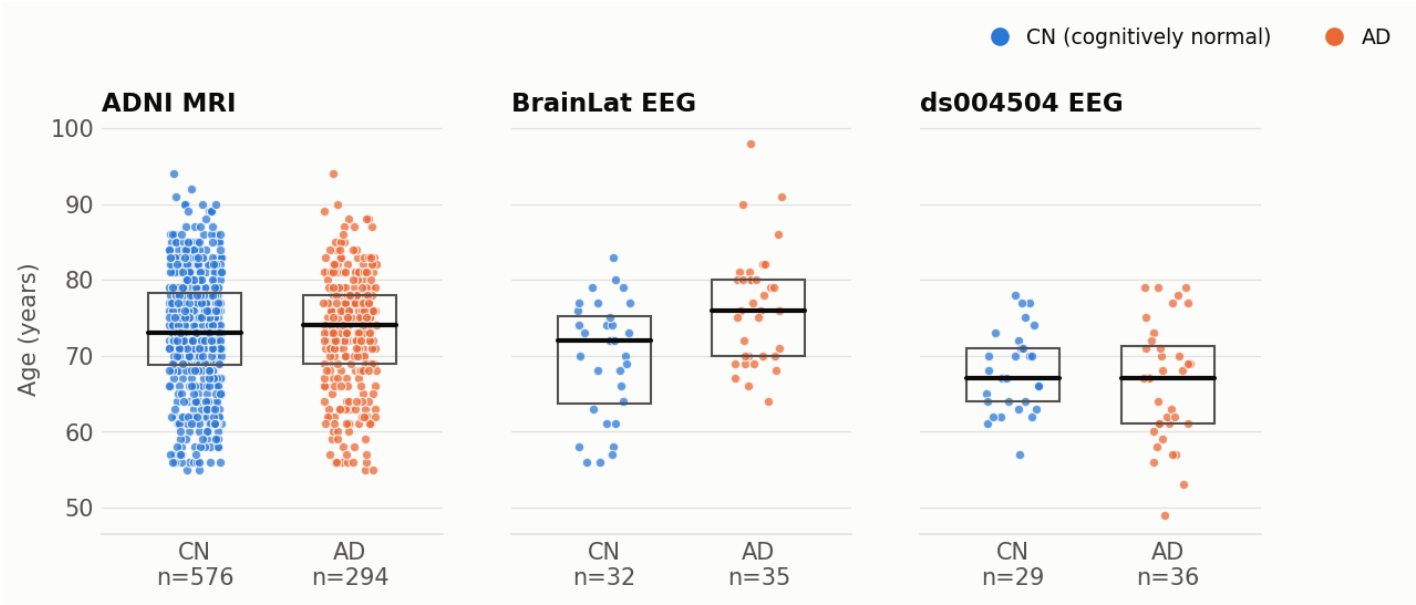


Figure 2. Age by diagnosis. Box = interquartile range, dark bar = median, dots = subjects.

3. MRI: ADNI structural measures

Source tables: `All_Subjects_UCSFFSX7` (UCSF FreeSurfer 7 cross-sectional runs), `DXSUM` (diagnosis), `PTDEMOG` (demographics), `DATADIC` and `Key_MRI`. One scan per subject (the earliest that passes QC), 3T only, diagnosis from the closest DXSUM visit (median gap 10 days). Each subject has 324 measures: cortical volume (_CV), surface area (_SA), thickness average (_TA) and SD (_TS), and subcortical volume (_SV).

- All 870 scans are 3T; FreeSurfer 7.2.0 (n=769), 7.4.1 (n=101).
- Only 101 runs carry a UCSF QC rating; the rest are unrated (`qc_rated = False`) but kept.
- Scanner make is known for 371 subjects (Siemens 256, GE 65, Philips 49, Other 1).
- Matching worked: age 73.2 vs 73.3 years, ~50% female in both groups. Education still differs (15.5 vs 16.6 years).

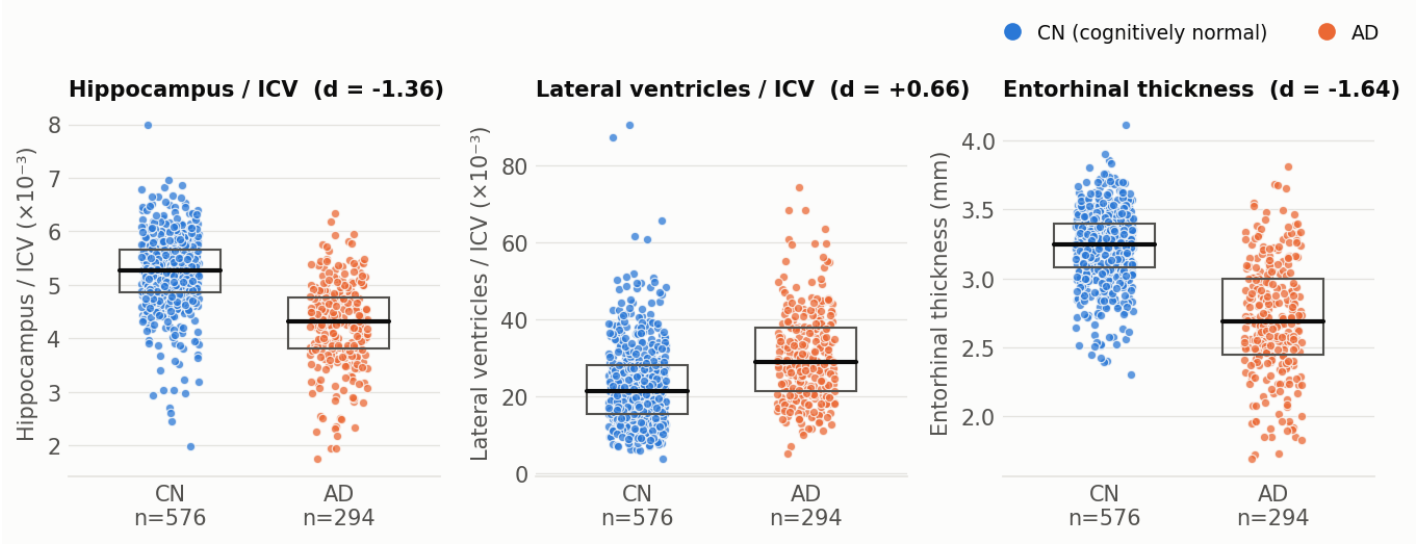


Figure 3. Classic AD atrophy markers: hippocampal and lateral-ventricle volume normalized to intracranial volume (ICV), and mean entorhinal cortical thickness. d = Cohen's d (AD − CN).

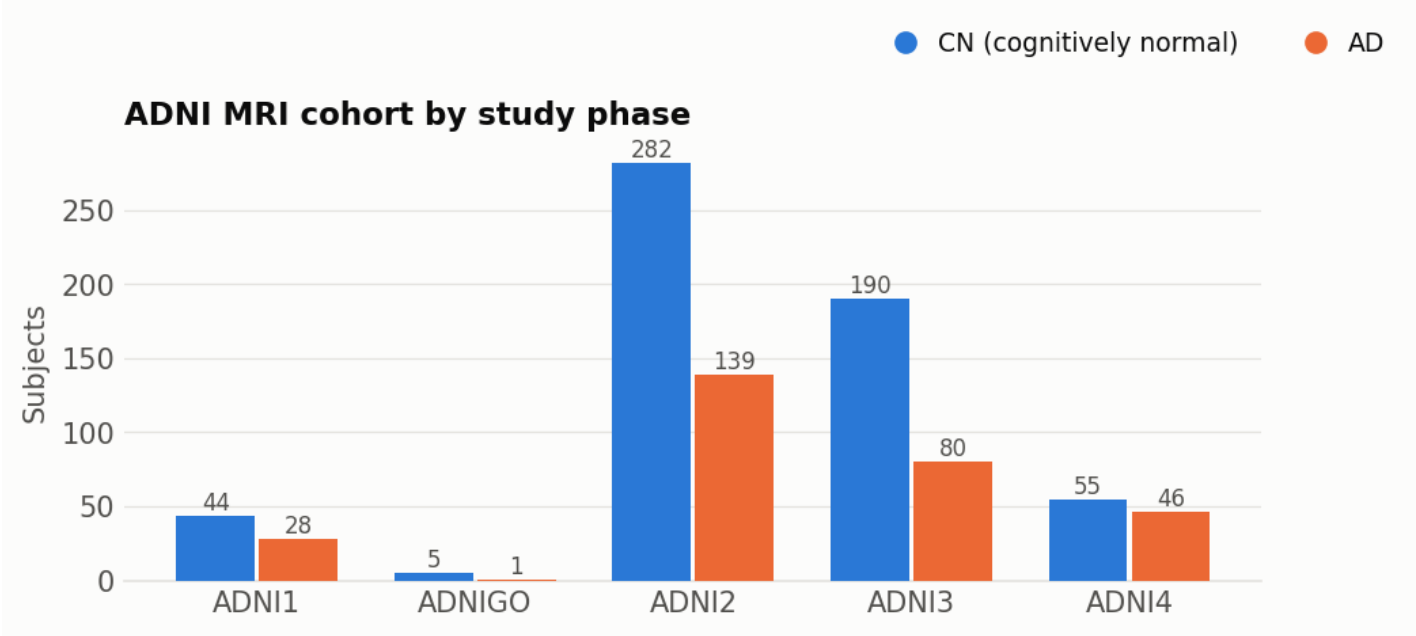


Figure 4. ADNI phase of each subject's scan. The AD:CN ratio varies by phase (CN were matched on age and sex, not phase), so phase/protocol should be a covariate.

4. EEG: BrainLat and ds004504

Both EEG datasets run through the same code (`prepare_brainlat_eeg.py` ; `prepare_ds004504_eeg.py` imports its functions): EEG channels only, average reference, 1–30 Hz band-pass, resample to 256 Hz, first ≤300 s. Features from Welch PSD (2 s windows, 50% overlap): relative power in delta (1–4 Hz), theta (4–8), alpha (8–13) and beta (13–30), globally and per region, theta/alpha ratio, and peak alpha frequency (7–13 Hz).

	BrainLat	ds004504
Recording	AD: `rs-HEP` files · CN: `rs` files	Eyes-closed rest (`task-eyesclosed`)
Amplifier / montage	BioSemi, 128 channels	Nihon Kohden, 19 channels (10–20; T3/T4/T5/T6 names)
Sampling rate	512 Hz (all)	500 Hz (all)
Sites	AR (Argentina) and CL (Chile)	Single site
Raw length, median	AD 424 s · CN 441 s	AD 829 s · CN 827 s
Files used	`s6_sub-*_rs[-HEP]_eeg.set` + .fdt	`sub-*/eeg/*_eeg.set` (raw, not derivatives)

Channels per scalp region

Regions come from 3-D head coordinates (front > 3 cm, back < –4 cm, lateral > 5 cm = temporal, otherwise central). The two montages differ a lot in density: a BrainLat region averages 19–44 channels, a ds004504 region 1–7. With this rule C3/C4 fall in "temporal", so ds004504 "central" is Cz alone and its central features are noisier than the others.

	Total	frontal	central	temporal	posterior
BrainLat (BioSemi 128)	128	35	19	30	44
ds004504 (10–20, 19 ch)	19	7	1	4	7

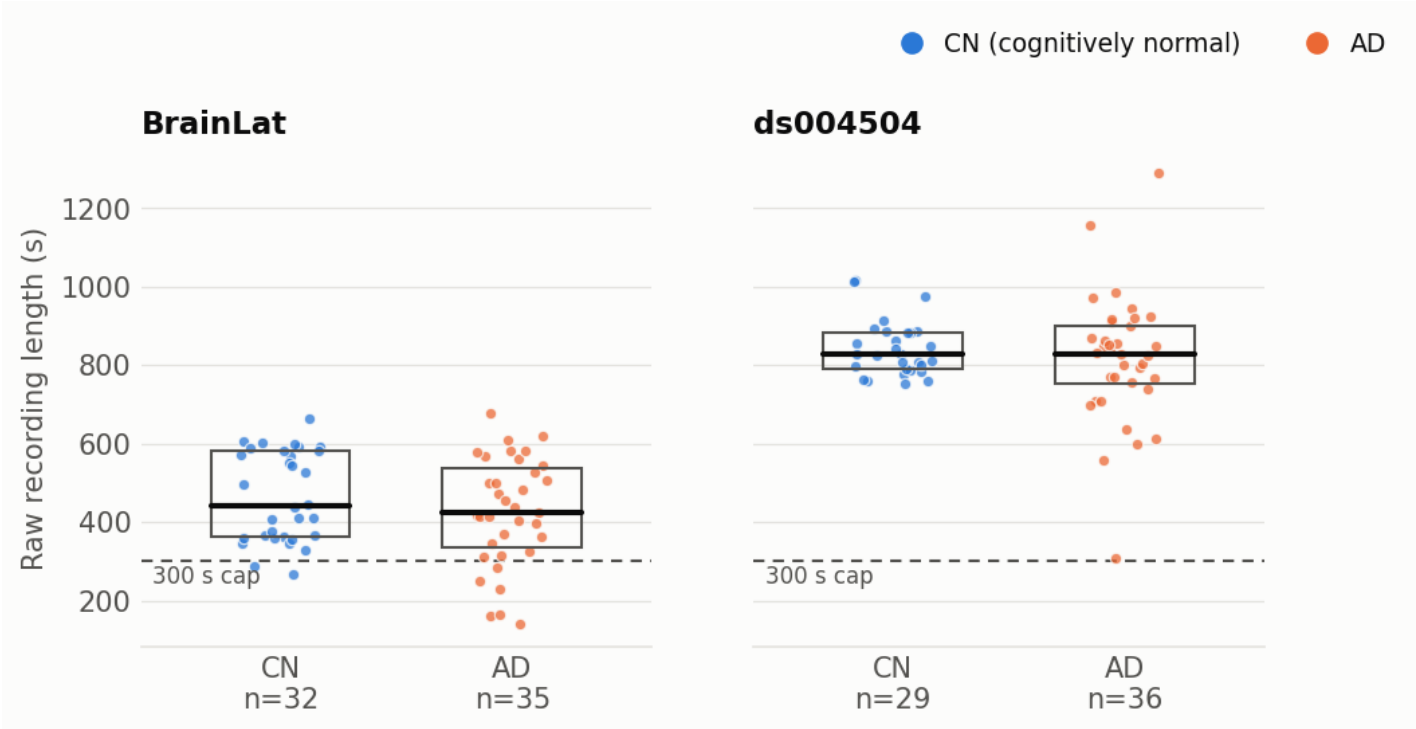


Figure 5. Raw recording length before the 300 s cap. 6 BrainLat AD recordings are shorter than 300 s (minimum 141 s), so those subjects contribute less data.

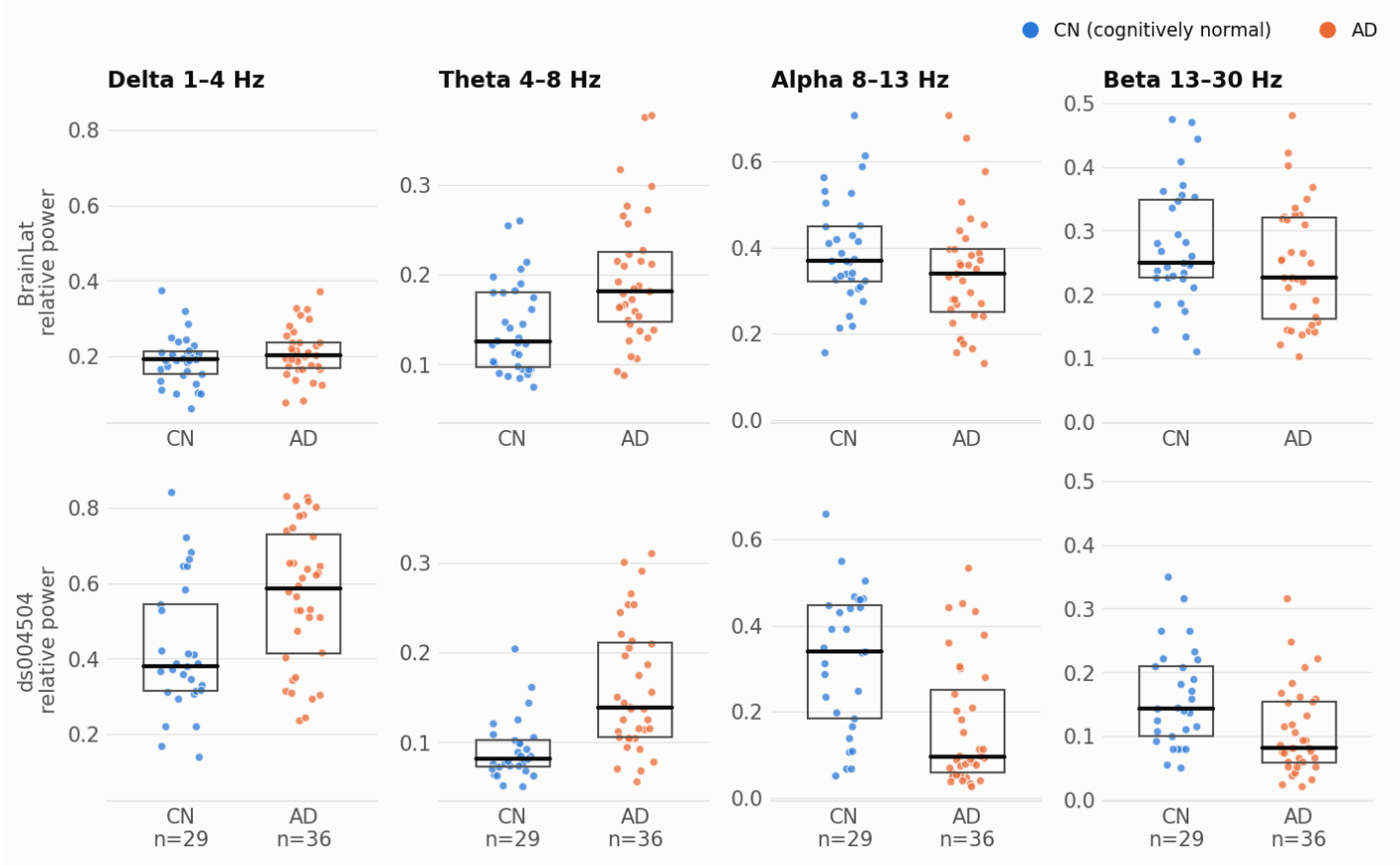


Figure 6. Global relative band power (fraction of 1–30 Hz power). Top row BrainLat, bottom row ds004504; each column shares its y-axis.

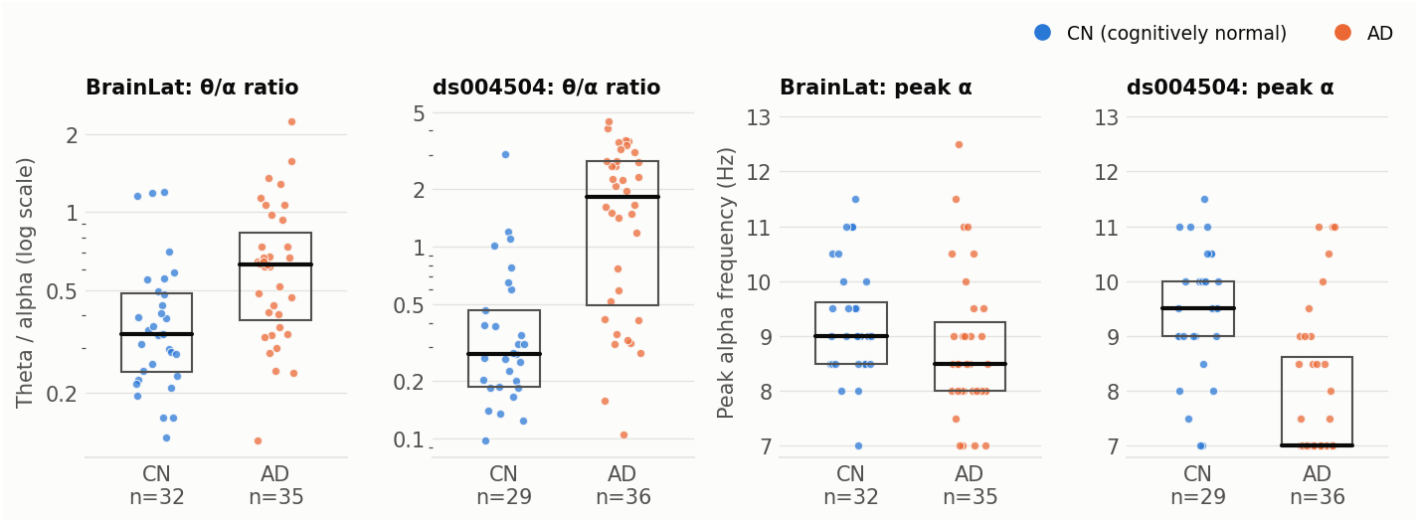


Figure 7. Theta/alpha ratio (log scale) and peak alpha frequency. AD shows EEG slowing in both datasets: higher θ/α and lower peak alpha. Peak alpha has 0.5 Hz resolution (2 s Welch windows) and is searched in 7–13 Hz, so values at 7 Hz (the ds004504 AD median) mean the spectral peak is at or below the search floor.

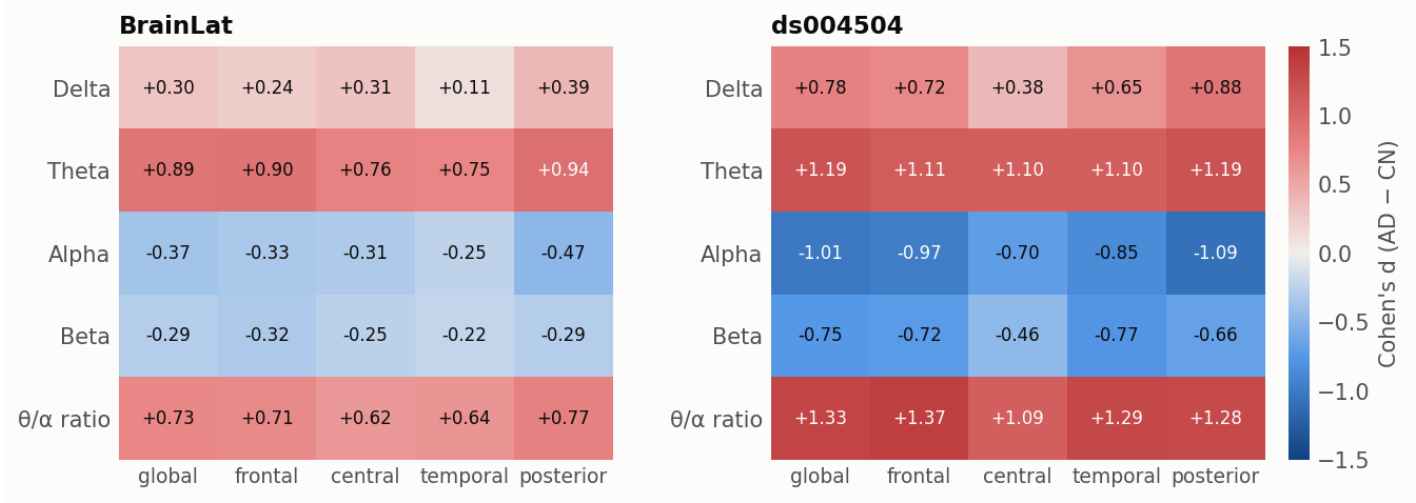


Figure 8. Effect size (Cohen's d, AD – CN) for each band and region. Red = higher in AD, blue = lower in AD.

Global EEG features: AD vs CN

Dataset	Feature	AD median	CN median	Cohen's d (AD–CN)	Mann–Whitney p
BrainLat	relpow_delta_global	0.20	0.19	0.30	0.226
BrainLat	relpow_theta_global	0.18	0.13	0.89	<0.001
BrainLat	relpow_alpha_global	0.34	0.37	-0.37	0.118
BrainLat	relpow_beta_global	0.23	0.25	-0.29	0.198
BrainLat	theta_alpha_ratio_global	0.63	0.34	0.73	0.001
BrainLat	peak_alpha_freq	8.50	9.00	-0.38	0.029
ds004504	relpow_delta_global	0.59	0.38	0.78	0.009
ds004504	relpow_theta_global	0.14	0.08	1.19	<0.001
ds004504	relpow_alpha_global	0.09	0.34	-1.01	<0.001
ds004504	relpow_beta_global	0.08	0.14	-0.75	0.003
ds004504	theta_alpha_ratio_global	1.80	0.28	1.33	<0.001
ds004504	peak_alpha_freq	7.00	9.50	-1.16	<0.001

5. Confounds and caveats

Issue	Dataset	Details	What to do
Recording type = diagnosis	BrainLat	All AD files are `rs-HEP`; all CN files are `rs`. Any AD/CN difference may partly reflect the recording protocol.	Confirm that rs-HEP is the same eyes-closed rest protocol; otherwise treat BrainLat results as exploratory and use ds004504 for replication.
Education	BrainLat, ADNI	BrainLat AD 9.3 vs CN 15.6 years; ADNI 15.5 vs 16.6. Not recorded in ds004504.	Covariate where available.
Age	BrainLat	AD 76.5 vs CN 69.8 years. Matched in ADNI and roughly similar in ds004504.	Covariate, or age-matched subset.
Sex	ds004504	AD 67% female vs CN 38%.	Covariate.
Recording length	BrainLat	6 AD recordings < 300 s (as short as 141 s); 2 CN < 300 s.	Sensitivity analysis at a common shorter cap (e.g. 140 s).
MMSE ceiling	ds004504	Every CN subject scores 30; MMSE is only available here.	Use only as an AD severity measure within AD.
Montage density	EEG, across datasets	128 vs 19 channels; region definitions are the same, but the electrodes behind them are not.	Prefer global features for cross-dataset comparison.
ADNI phase / QC	ADNI	AD:CN ratio varies by phase; 769 of 870 FreeSurfer runs are not QC-rated.	Phase as covariate; check that results hold in the QC-rated subset.
Not the same subjects	All	EEG and MRI come from different cohorts (no subject has both).	Cross-modal analyses compare group-level effects, not paired data.

6. Files

File	Contents
`processed/mri/adni_mri_cohort.csv`	One row per ADNI subject: IDs, dx, scan info, demographics, 324 FreeSurfer measures
`processed/eeg_full/eeg_features.csv`	BrainLat AD/CN EEG features (one row per subject)
`processed/eeg_full/eeg_inventory.csv`, `eeg_missing_subjects.csv`	Status of every BrainLat .set file; subjects without usable data
`processed/ds004504/eeg_features.csv`	ds004504 AD/CN EEG features, same columns as BrainLat plus `mmse`
`processed/eeg/brainlat_verification.csv`, `processed/ds004504_verification.csv`	Download checks (file present, size, loads, sfreq, channels, duration)
`processed/eeg/synapse_ad_search.csv`	Every AD file on BrainLat Synapse (metadata)